PCoA biplots & pathogen screening

PCoA biplots

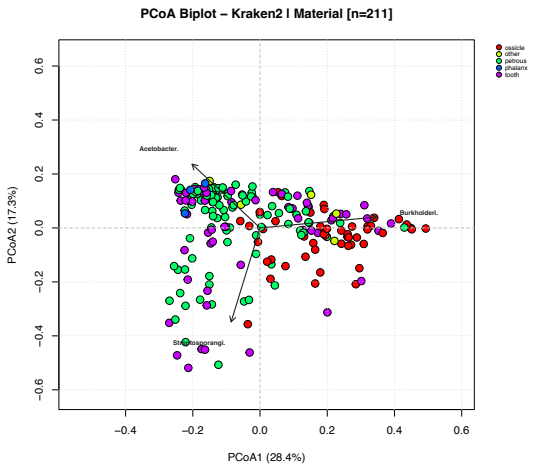
I added biplot arrows to all 10 PCoA plots to show which bacteria families are most responsible for the differences we see between samples. These arrows represent the strongest statistical 'drivers' of the groups.

The results are consistent across both tools:

Tool	Family	r^2
Kraken2	Streptosporangiaceae	0,922
Kraken2	Burkholderiaceae	0,873
Kraken2	Acetobacteraceae	0,693
KrakenUniq	Burkholderiaceae	0,968
KrakenUniq	Rhodobacteraceae	0,742
KrakenUniq	Streptosporangiaceae	0,659

All top drivers are soil bacteria :(

This confirms that the differences are due to the burial environment (soil bacteria absorbed by the bones over time) rather than the original biology of the individuals. The two tools agree on the main drivers (*Burkholderiaceae* and *Streptosporangiaceae* in both), with minor differences in secondary families reflecting the different databases (GTDB vs MicrobialNT).
The plots show that certain bone types, like the ossicles (red points), cluster together because they were more exposed to soil bacteria like *Burkholderiaceae*. This suggests the separation is a matter of bone porosity and soil absorption, not biological differences between people.



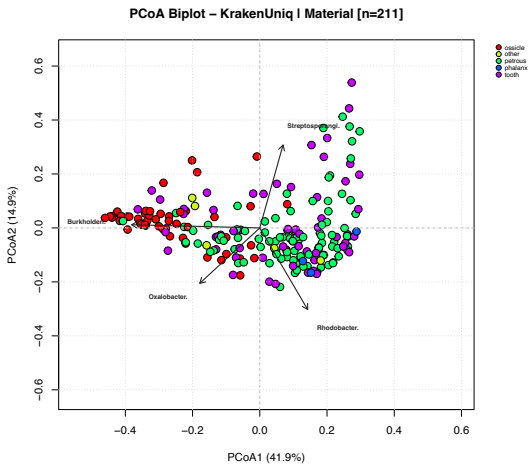
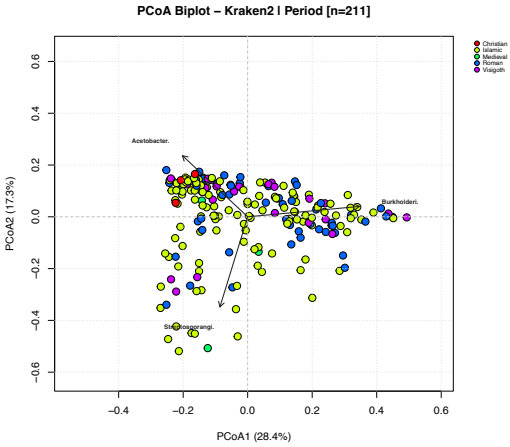


Fig. 1 (left): PCoA Biplot, Kraken2 | Material (n=211). Arrows show top envfit family drivers, scaled by r^2. *Burkholderiaceae* (r^2=0.873) points toward the ossicle cluster along PC1; *Streptosporangiaceae* (r^2=0.922) drives PC2 downward.
Fig. 2 (right): PCoA Biplot, KrakenUniq | Material (n=211). *Burkholderiaceae* dominant driver (r^2=0.968), pointing toward the ossicle cluster. Four arrows shown vs. three in Kraken2, reflecting the broader taxonomic coverage of MicrobialNT.



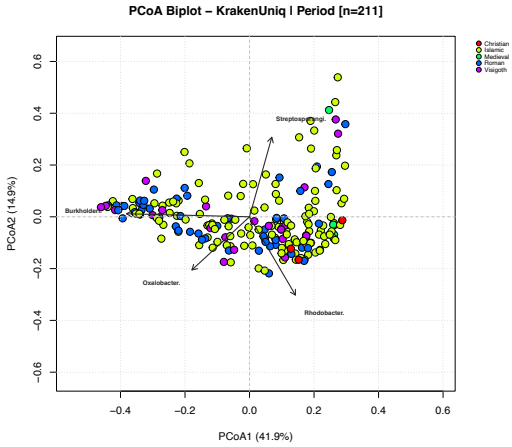
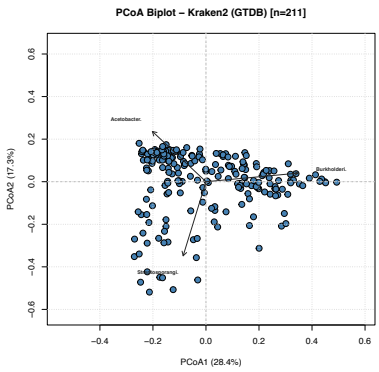
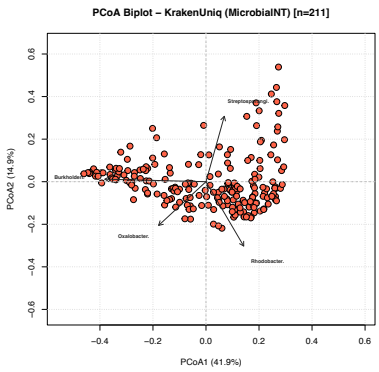
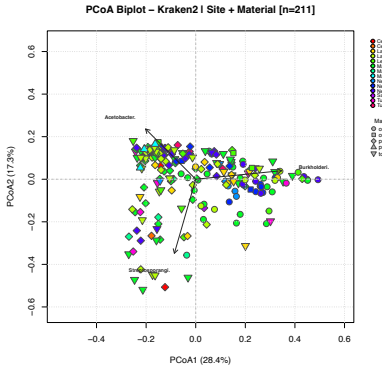


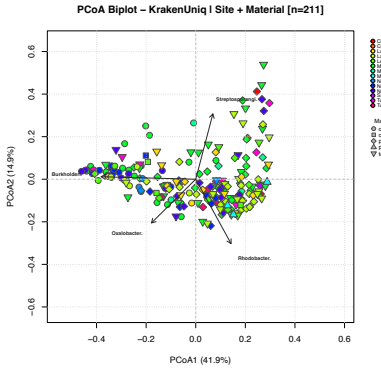
Fig. 3 (left): PCoA Biplot, Kraken2 | Period (n=211). No clustering by historical period is visible, all five periods are intermixed across the ordination space, confirming that the dominant signal is taphonomic rather than cultural.
Fig. 4 (right): PCoA Biplot, KrakenUniq | Period (n=211). Same result as Kraken2: no period-based structure. The Islamic samples (yellow-green, numerically dominant) are distributed throughout the ordination space.

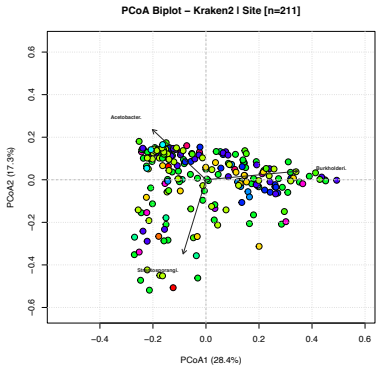
Other plots

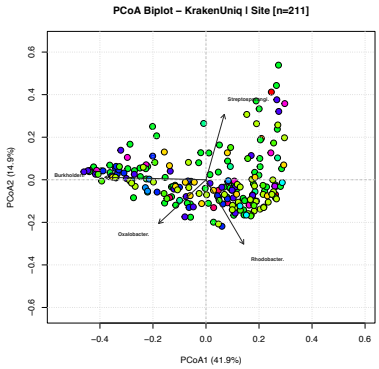












Pathogen screening (KrakenUniq)

I screened all 250 KrakenUniq reports on Dardel for three pathogens, extracting per-sample clade reads, direct reads, unique k-mers, and coverage at species level.
To confirm if a pathogen is real and not just a 'false positive' from soil bacteria, I used a 'specificity score' (unique k-mers). Think of it as a genetic fingerprint that belongs *only* to that pathogen. I used a strict threshold to ensure the results are authentic.
I used KrakenUniq rather than Kraken2 because the unique k-mer column exists only in KrakenUniq reports; Kraken2 counts total k-mer matches but does not track how many distinct genomic positions they cover, making it unable to distinguish authentic signal from cross-mapping.

Results after filtering to the project samples in the metadata

- **Yersinia pestis.** 108 samples have at least 1 read assigned to *Y. pestis*. The maximum unique k-mers in any project sample is **20** (P32015_1038, Medieval, Cementerio de San Lorenzo). The two samples with the highest combined signal are both Roman, from Legionarios Sertorianos, L'Almoina (P35761_1087: 154 direct reads, 16 k-mers; P35761_1091: 52 reads, 18 k-mers): historically interesting, but well below any authentication threshold. No authentic signal found; the few reads detected are just noise from environmental relatives (like soil bacteria).
- **Tannerella forsythia.** 112 samples have at least 1 read assigned to *T. forsythia*. Five project samples exceed the ≥1000 unique k-mers threshold. The two tooth samples (both Islamic period, La Rauda, L'Almoina) confirm that *T. forsythia* is present in the dental samples with a strong, authentic-looking signal (>200 direct reads, well above the standard quality thresholds for ancient DNA). One finger bone (phalanx) showed a high signal for this oral pathogen. This is likely due to contamination from dental material in the same burial, or potentially a severe infection that spread through the blood. My conclusion: *T. forsythia* is genuinely detectable in the Valencia dataset, particularly in Islamic-period dental samples from L'Almoina.
- **Mycobacterium tuberculosis (my idea).** 188 samples have at least one read assigned to *M. tuberculosis*. The maximum unique k-mers across all project samples is **334**, with most top candidates showing 0 direct reads despite having clade reads: the classic signature of cross-mapping from environmental soil *Mycobacteriaceae*. No sample approaches the ≥1000 threshold, and the distribution of materials and periods shows no recognisable biological pattern. My conclusion: no authentic *M. tuberculosis* signal detectable in the Valencia dataset.

NGI_ID	Direct reads	Unique k-mers	Period	Material
P35761_1113	2439	42707	Christian	phalanx
P35503_1106	715	12657	Islamic	tooth
P32015_1009	562	8372	Roman	petrous
P35761_1102	261	3546	Islamic	tooth
P35503_1091	83	1193	Islamic	petrous

Summary across all three pathogens

T. forsythia is the only pathogen with an authentic detectable signal, concentrated in Islamic-period dental samples from L'Almoina.

Pathogen	Max unique k-mers	Verdict
Yersinia pestis	20	Background noise
Mycobacterium tuberculosis	334	Background noise
Tannerella forsythia	42707	True signal